Conserved Yeast Heat-shock Response in Human and Mouse Cells

Bruno Truzzi

January 2026

Abstract

Cellular stress responses allow organisms to maintain homeostasis under adverse environmental conditions such as heat shock. In this project, we analyze heat-shock–induced transcriptional responses using RNA-sequencing data from yeast, mouse, and human cells. Differential analysis of gene expression was combined with homology mapping and functional enrichment to assess conservation at both the sequence and functional levels. Our analysis reveals that core stress-response processes, such as protein-folding-related ones, are conserved across species, while species-specific differences are also observed. This shows not only evolutionary conservation, but also the divergence in transcriptional stress responses across eukaryotes, even among homologous genes.

1 Introduction

Cellular stress responses are important mechanisms which allow organisms to adapt to environmental changes. Heat stress, for example, disrupts protein folding, transcription, and other cellular processes. Therefore, organisms need a response to maintain cellular integrity. A central component of this response is the regulation of gene expression.

Comparing stress-induced gene expression reactions across species can allow us to investigate how conserved these responses are and also to identify regulatory mechanisms which have been preserved through evolution. This comparison can also reveal adaptations that are species-specific and also show differences in cellular complexity.

In this project, we analyze heat-shock–induced transcriptional responses using RNA sequencing data from yeast, mouse, and human cells. By using differential expression analysis with homology mapping and functional enrichment, we aim to assess the degree of conservation of stress responses across eukaryotes. In particular, we focus on whether genes that are conserved at the sequence level also show shared functional roles under heat-shock conditions.

2 Methods

2.1 Data sources

All data (apart from yeast complex heat-shock) used were obtained from publicly available RNA-sequencing datasets deposited in the Gene Expression Omnibus (GEO). Only samples associated with heat-shock conditions and their corresponding controls were included in the analysis. Below is a summary of the datasets.

HeLa cell data was taken from the GEO series GSE247417[14]. This dataset consists of RNA-sequencing measurements from HeLa cells exposed to heat shock and untreated controls, with three biological replicates per condition. Raw read counts were provided. The series metadata indicate heat-shock treatment but do not specify the temperature or duration of the heat-shock exposure

Additionally, human heat-shock data was taken from the GEO series GSE100469[7], which contains RNA-sequencing data from primary human foreskin fibroblasts. Cells were exposed to heat shock at 44 °C, and RNA was collected in two time windows (0–1 h and 1–2 h after heat-shock onset), together with unstressed control samples. Two biological replicates were available for each condition. Expression values were provided in FPKM measurements.

Mouse data was obtained from the GEO series GSE133251[10], generated from mouse melanoma cells. Cells were subjected to heat shock at 45 °C for 1 hour, The dataset includes heat-shocked and control samples, with three biological replicates per condition. Expression values were provided in the form of FPKM measurements.

2.2 Differential Expression Analysis

2.2.1 Performing DESeq2 on Yeast and HeLa Data

Differential expression analysis was performed using DESeq2[9] for the yeast and HeLa datasets, which were available as raw gene-level count data.

For the yeast data, the analysis was restricted to wild-type samples. Heat-shock responses were assessed by comparing expression at 37 °C after 10 minutes to the baseline condition (25 °C, 0 minutes). To reduce the influence of spikes in expression, genes identified as differentially expressed at 10 minutes were further required to show the same direction of regulation at 37 °C after 30 minutes.

For the HeLa dataset, differential expression was assessed by comparing heat-shocked samples to untreated control samples. Log2 fold change shrinkage was used for the HeLa dataset.

In both datasets, genes were considered differentially expressed if they showed an adjusted p-value below 0.05 and an absolute log2 fold change greater than 1.

2.2.2 Performing limma on Fibroblast and Mouse Data

Differential expression analysis was performed using limma[12] for the mouse and human fibroblast datasets, which were available as normalized expression values (FPKM) instead of raw counts. limma was used as a alternative to DESeq2 for DEA.

For the mouse dataset, gene expression values were log-transformed prior to analysis. Differential expression was assessed by comparing heat-shocked samples to non-treated control samples using a linear model with condition as the explanatory variable. Variance estimates were moderated using empirical Bayes methods. Genes were classified as differentially expressed if they showed an adjusted p-value below 0.05 and an absolute log2 fold change greater than 0.5.

For the human fibroblast dataset, normalized gene expression values were log-transformed and additionally subjected to between-array normalization to improve comparability across samples. Differential expression was assessed by comparing heat-shocked samples to mock-treated controls. Genes were considered differentially expressed using the same statistical thresholds of an adjusted p-value <0.05 and an absolute log2 fold change >0.5.

A lower log2 fold change threshold was used for the limma-based analyses to account for the narrower distribution of observed effect sizes in these datasets. In the limma analyses, 90% of log2 fold change values were roughly between -1.2 and $+2.4$. In contrast, the DESeq2 analysis showed a wider distribution, with 90% of values between approximately -2.6 to $+5.0$.

2.3 HomoloGene Mapping

To compare heat-shock responses across species, we mapped the human (HeLa, fibroblast) and mouse DEG sets to their yeast homologous genes using HomoloGene[11]. HomoloGene uses Entrez Gene IDs, so all DEG lists were first converted to Entrez IDs using organism-specific annotation packages.

Yeast DEGs were mapped using org.Sc.sgd.db[4].

Human DEGs (HeLa and fibroblasts) were mapped from Ensembl gene IDs to Entrez using org.Hs.eg.db[3]. Mouse DEGs were filtered to Ensembl gene IDs and mapped to Entrez using org.Mm.eg.db.

Homology was established by identifying HomoloGene IDs (HIDs) that contain at least one DEG from each species in the comparison. For each HID, we recorded how many distinct genes from each species were present and aggregated the corresponding Entrez IDs.

Homology mapping was performed separately for HeLa–yeast and fibroblast–yeast. In addition, a combined human DEG set was created by taking the union of HeLa and fibroblast DEGs, and human–yeast homolog mapping was recomputed on this unified set to capture homologous relations present in both human cell type.

2.4 Enrichment

Functional enrichment analysis was performed using clusterProfiler[16] to identify Gene Ontology biological processes associated with conserved heat-shock responses across species.

GO enrichment was used separately for mouse–yeast and human–yeast homologous DEG sets. Foreground gene lists were defined from the HomoloGene mappings and consisted of Entrez Gene IDs for genes that were differentially expressed in one species and had a mapped homolog in yeast. Backgrond sets consisted from all genes present in the corresponding expression datasets that could be mapped to Entrez IDs. Foreground DEG sets were intersected with their background.

GO enrichment was performed using the Biological Process ontology with Benjamini–Hochberg correction for multiple testing. Terms were considered enriched using a p-value cutoff of 0.01 and a q-value cutoff of 0.2. Enrichment was computed independently for each species, and conserved functional responses were identified by intersecting significantly enriched GO terms between species based on shared GO identifiers.

To explore relationships between enriched GO terms, a semantic similarity network was constructed using GOSemSim[15], igraph[5] and GO.db[2]. Semantic similarity between GO Biological Process terms was computed using a graph-based measure, and GO terms were connected if their similarity exceeded a predefined threshold. The resulting network integrates enriched terms from human and yeast

3 Results

3.1 Differential Expression Analysis

In yeast, a total of 2,579 genes were identified as significantly differentially expressed. The HeLa dataset showed 5,044 genes, while the human fibroblast dataset yielded 1,130 differentially expressed genes. In mouse, 3,485 genes.

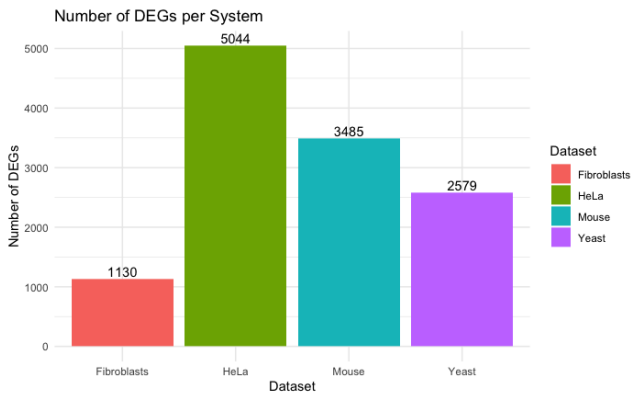

Figure 1: Generated in R

3.2 Homologene Mapping

Individually, the HeLa data showed 55 homolog mappings, while the fibroblast data showed 47 mappings. For the combined human DEG set (union of HeLa and fibroblasts), a total of 97 human–yeast homolog mappings were obtained. For the mouse dataset, 121 mouse–yeast homolog mappings were identified.

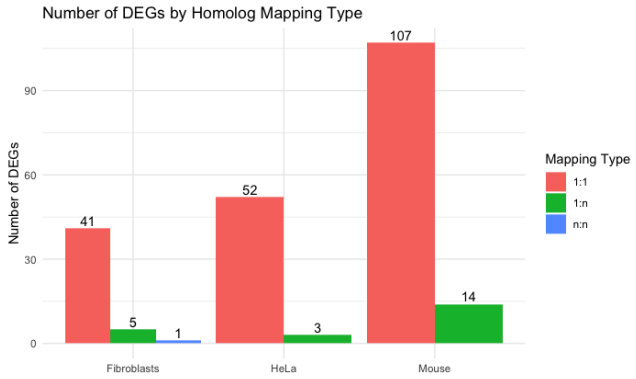

Figure 2: Generated in R

3.3 Enrichment

For the human–yeast comparison (using the union of HeLa and fibroblast DEGs), 54 enriched GO terms were identified. Among these, 8 GO terms were shared between human and yeast, while 42 terms were specific to human and 4 terms were specific to yeast. Most significant shared terms included protein folding, ribosome biogenesis and tRNA- related terms.

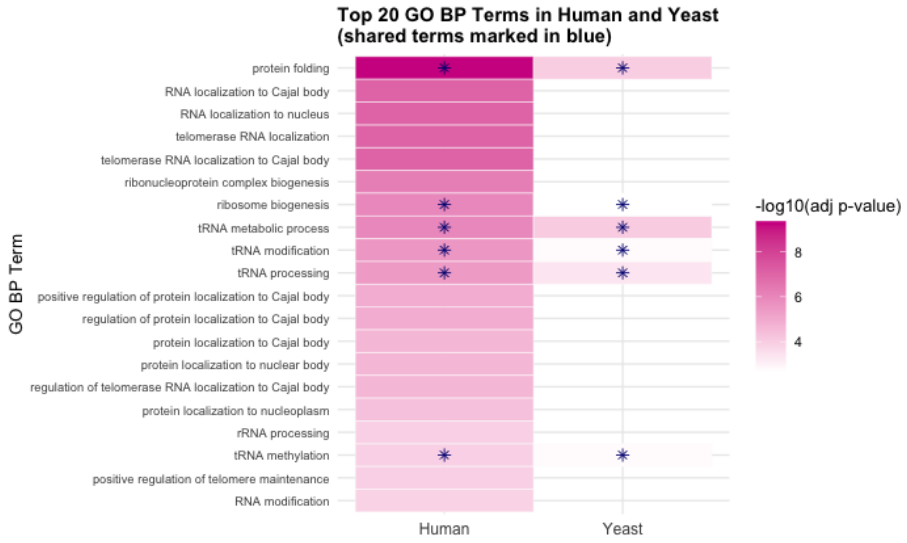

Figure 3: Generated in R

For the mouse–yeast comparison, enrichment analysis identified 29 significantly enriched GO terms in total. Of these, 12 GO terms were shared between mouse and yeast, while 15 terms were specific to mouse and 2 terms were specific to yeast. Most significant shared terms included tRNA and rRNA related terms and ribosome biogenesis.

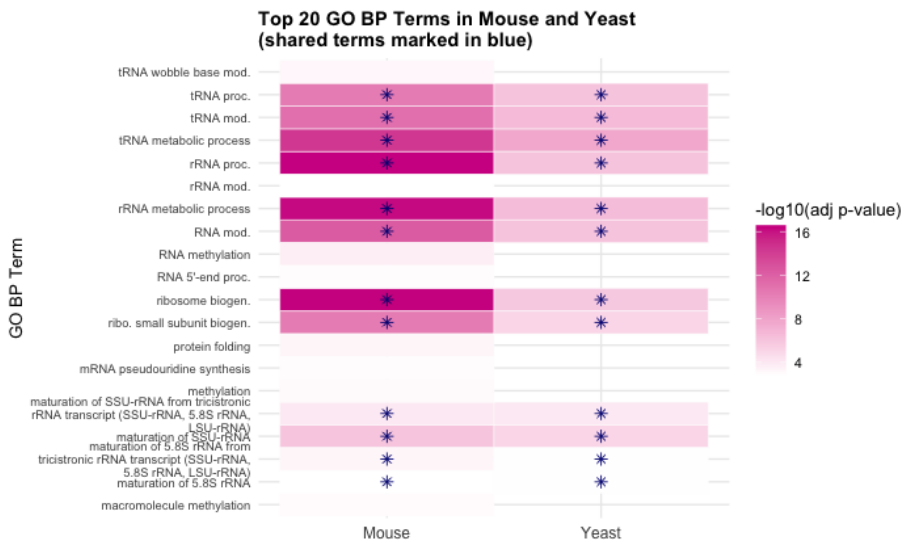

Figure 4: Generated in R

4 Discussion

The analysis of the heat shock response in yeast and human cells revealed both conserved and species-specific biological processes. Differential expression analysis showed that heat shock caused changes in gene expression in all datasets. However, homology mapping revealed that only a small fraction of these differentially expressed genes was shared between yeast and humans/mice. For Mice, approximately 3.2% of all DEGs could be mapped to a yeast homologous gene. For the combined human DEG set, approximately 1.7% had a yeast homologue.

Most homology relationships between human cells and yeast were 1:1 homologous. The same was true for mice. For humans, only 9 out of the total 97 mapped genes were not 1:1 mappings. For mice, only 14 out of 107. Yeast and mammals diverged more than 1 billion years ago [8]. As a result, many genes that are still conserved between these species represent essential core cellular functions. Consequently, these conserved genes tend to remain as single-copy genes in both species.

Even though the number of successfully mapped homologous DEGs between organisms was low, GO enrichment analysis identified several biological processes . The most significantly enriched GO terms were related to protein folding. This is biologically expected, as heat stress leads to protein unfolding, which requires activation of protein folding [13]. The presence of these GO terms in both species indicates that the core mechanisms protecting proteins from heat damage are evolutionarily conserved.

In addition to protein folding, several shared GO terms were related to translation, including tRNA modification, tRNA methylation, and tRNA processing. These processes are important for controlling protein synthesis under stress conditions [6]. Their presence in yeast and human cells suggests that translation is another conserved response to heat shock.

Although protein folding and translation related processes were shared, they were much more significant processes in human cells than in yeast. These human-specific terms were mainly associated with RNA-related and ribosome-related processes. This could indicate that RNA processing and RNA localization are more extensively regulated in human cells than in yeast during heat stress. This interpretation is consistent with the higher proteomic and regulatory complexity of human cells, which devote a larger fraction of their proteome to regulatory functions compared to yeast [1]. Mouse data also showed similar shared GO terms with yeast (particularly tRNA-related terms) and substantially more specific significant terms, which resonates with the human results.

Additionally, GO terms related to reproduction and cell–cell recognition were found only in human cells, since these processes do not exist in yeast. Furthermore, despite restricting the analysis to genes with identifiable homologous mappings between species, a substantial number of enriched biological processes remained specific to human cells. This indicates that the presence of a homologous gene alone does not necessarily imply functional equivalence under a given condition. Even when sequence-level similarity is conserved, genes may participate in different biological processes.

Overall, the results show that the basic heat shock response is particularly conserved between yeast and humans for protein folding and translation-related processes. However, human cells display a much more complex and detailed transcriptional response. This shows how fundamental stress responses are preserved through evolution, while organism specific functions and regulatory complexity have diverged.

A few limitations should be noted. Only a small fraction of DEGs were conserved between yeast and human cells, which limits the number of genes available for cross-species comparison and reduces the the ability to detect meaningful differences. Additionally, different differential expression methods were used across datasets due to partial normalization, which may influence the results. Despite these limitations, the conserved processes likely represent core components of the heat shock response.

References

[1] Martin Beck, Alexander Schmidt, Johan Malmström, Manfred Claassen, Alessandro Ori, Anna Szymborska, Franz Herzog, Oliver Rinner, Jan Ellenberg, and Ruedi Aebersold. The quantitative proteome of a human cell line. *Molecular Systems Biology*, 7:549, 2011.

[2] Marc Carlson. *GO.db: A set of annotation maps describing the entire Gene Ontology*, 2025. R package version 3.22.0.

[3] Marc Carlson. *org.Hs.eg.db: Genome wide annotation for Human*, 2025. R package version 3.22.0.

[4] Marc Carlson. *org.Sc.sgd.db: Genome wide annotation for Yeast*, 2025. R package version 3.22.0.

[5] Gábor Csárdi and Tamás Nepusz. The igraph software package for complex network research. *InterJournal*, Complex Systems:1695, 2006.

[6] Chen Gu, Thomas J. Begley, and Peter C. Dedon. trna modifications regulate translation during cellular stress. *FEBS Letters*, 588(23):4287–4296, 2014.

[7] Hennig T, Michalski M, Rutkowski A, Djakovic L, Whisnant A, Friedl M, Jha B, Baptista M, L'Hernault A, Erhard F, Dölken L, Friedel CC. 4su-seq of hff exposed to salt and heat stress. Gene Expression Omnibus, 2018. Accessed via GEO accession GSE100469.

[8] Jon M Laurent, Riddhiman K Garge, Ashley I Teufel, Claus O Wilke, Aashiq H Kachroo, and Edward M Marcotte. Humanization of yeast genes with multiple human orthologs reveals functional divergence between paralogs. *PLoS Biology*, 18(5):e3000627, 2020.

[9] Michael I. Love, Wolfgang Huber, and Simon Anders. Moderated estimation of fold change and dispersion for rna-seq data with deseq2. *Genome Biology*, 15:550, 2014.

[10] Luo M, Shao B. The effect of yap/taz on the expression of the heat shock transcriptome. Gene Expression Omnibus, 2020. Accessed via GEO accession GSE133251.

[11] Ogan Mancarci. *homologene: Quick Access to Homologene and Gene Annotation Updates*, 2019. R package version 1.4.68.19.3.27.

[12] Matthew E Ritchie, Belinda Phipson, Di Wu, Yifang Hu, Charity W Law, Wei Shi, and Gordon K Smyth. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*, 43(7):e47, 2015.

[13] Jimmy Verghese, Jesse Abrams, Yong Wang, and Kevin A. Morano. Biology of the heat shock response and protein chaperones. *Cell Stress & Chaperones*, 17(2):153–166, 2012.

[14] Wang W, Wang B. Effect of clk2 expression in the regulation of intron retention during heat shock. Gene Expression Omnibus, 2023. Accessed via GEO accession GSE247417.

[15] Guangchuang Yu. Gene ontology semantic similarity analysis using gosemsim. *Methods in Molecular Biology*, 2117:207–215, 2020.

[16] Guangchuang Yu, Li-Gen Wang, Yanyan Han, and Qing-Yu He. clusterprofiler: an r package for comparing biological themes among gene clusters. *OMICS: A Journal of Integrative Biology*, 16(5):284–287, 2012.